

BASHIRU SANI

Microbiologist | PhD Researcher | Lecturer

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PROFESSIONAL PROFILE

Microbiologist and PhD researcher with a multidisciplinary background spanning medical microbiology, environmental microbiology, microbial ecology, antimicrobial resistance, bioremediation and omics. Current doctoral research examines amoebae-microbe interactions in hydrocarbon-impacted environments and their implications for remediation and antimicrobial resistance. University lecturer and peer reviewer with experience in molecular microbiology, sequencing-based research, scientific data analysis, teaching, student mentoring and research communication. Career focus: research and applied roles in environmental microbiology, bioremediation, One Health, microbial ecology and omics-enabled environmental science.

CORE EXPERTISE

- Environmental microbiology & microbial ecology
- Hydrocarbon bioremediation
- Multi-omics, metagenomics & bioinformatics
- Scientific writing, peer review & research synthesis
- Medical microbiology & mycology
- AMR & One Health
- Molecular microbiology: PCR/qPCR, DNA & sequencing
- University teaching, supervision & mentoring

RESEARCH & ACADEMIC EXPERIENCE

University of Strathclyde | Glasgow, UK

2024 - Present

PhD Researcher, Civil & Environmental Engineering

- Investigate the roles of free-living amoebae and microbial symbionts in hydrocarbon remediation and antimicrobial resistance dynamics using omics-based approaches.
- Work across environmental microbiology, microbial ecology and molecular methods, including DNA extraction, qPCR, microscopy and sequencing-based microbial community analysis.
- Interpret complex biological and environmental datasets to identify microbial patterns, interactions and candidate functional pathways relevant to contaminated-site remediation.
- Develop research outputs including reviews, conference presentations, figures, data summaries and publication-ready scientific writing.

Federal University of Lafia | Lafia, Nigeria

Aug 2023 - Present

Assistant Lecturer, Department of Microbiology

- Teach and support undergraduate microbiology across general microbiology, molecular techniques, petroleum microbiology and water/sewage microbiology.
- Conduct microbiological and molecular research, coordinate laboratory sessions, analyse research data and contribute to manuscripts and grant-related activities.
- Mentor and supervise students in research design, scientific writing, presentations and publication development; 40+ students mentored across academic activities.
- Contribute to departmental research and e-learning activities and to curriculum development, including bioinformatics-related teaching content.

Federal University of Lafia | Lafia, Nigeria

Aug 2018 - Aug 2023

Graduate Assistant, Department of Microbiology

- Supported teaching, practical classes, assessment, student guidance and laboratory research while progressing through postgraduate training.

Graduate Researcher, Biotechnology

- Characterised the *Trypanosoma cruzi* NB2 transporter gene using bioinformatics, DNA sequencing/alignment, primer design, cloning and phylogenetic analysis.
- Applied CRISPR-Cas9 mutagenesis and laboratory methods including PCR/qPCR, gel electrophoresis, Western blotting, protein expression, growth analysis and drug sensitivity assays.

Biotechnology Advanced Laboratories, SHETSCO | Abuja, Nigeria

2014

Research Assistant

- Supported antimicrobial assays, plasmid purification, bacterial/plant DNA extraction, plant tissue culture, fermentation assays and prodigiosin purification.

EDUCATION

University of Strathclyde, UK - PhD, Civil & Environmental Engineering

2024 - Present

Research focus: amoebae and microbial symbionts in hydrocarbon remediation and antimicrobial resistance dynamics using omics approaches.

Federal University of Lafia, Nigeria - MSc Microbiology (Medical)

2023

Distinction - CGPA 4.59/5.00

Thesis: Genetic Diversity and Antimicrobial Susceptibility of Mucor, Candida and Aspergillus from COVID-19 First Responders in Lafia, Nasarawa State.

University of Glasgow, UK - PGDip Biotechnology

2019

Research project: Assessment and Characterisation of Trypanosoma cruzi NB2 (TcrNB2) Transporter Gene.

Federal University of Lafia, Nigeria - BSc (Hons) Microbiology

2015

First Class - CGPA 4.66/5.00

Project: Prevalence and Antimicrobial Susceptibility of Listeria monocytogenes from Fresh Cow and Goat Meat Sold in Lafia.

TECHNICAL SKILLS

Molecular & microbiology: microbial culture; DNA extraction; PCR/qPCR; gel electrophoresis; antimicrobial susceptibility testing; microscopy; cloning; Western blotting.

Omics & bioinformatics: 16S/ITS/18S sequencing workflows; metagenomics concepts; sequence alignment; BLAST; phylogenetics; biological data interpretation.

Environmental analysis: hydrocarbon-remediation experimental design; GC-FID/TPH workflow exposure; environmental sample processing and microbial community analysis.

Data & computing: IBM SPSS; Microsoft Excel; Linux; R and Python (developing/basic proficiency).

Research communication: scientific writing; literature review; peer review; bibliometric/scientometric analysis; conference presentation; grant and research proposal development.

SELECTED PUBLICATIONS

Selected peer-reviewed outputs; full publication profile available on Google Scholar.

- Yusuf, M., Mkpouto, P., Abduljalal, Y., Bashiru, S., et al. (2021). SARS-CoV-2 Spike Protein and RNA Dependent RNA Polymerase as Targets for Drug and Vaccine Development: A Review. Biosafety and Health, 3(5), 249-263. <https://doi.org/10.1016/j.bsheal.2021.07.003>
- Bashiru, S.E., Femi, G., Kabiru, S., Naja'atu, H.S. & Fatima, A.Y. (2022). Prevalence and antibiogram profile of Listeria monocytogenes associated with cattle farms in Lafia, Nasarawa State. Bacterial Empire, e475. <https://doi.org/10.36547/be.475>

- Peter, U.U., Bashiru, S., Naja'atu, S.H., Fatima, Y.A. & Kabiru, S. (2022). Molecular Pathogenesis of Rocky Mountain Spotted Fever: A Brief Review. IMC Journal of Medical Science, 16(1), 004. <https://doi.org/10.55010/imcjms.16.010>
- Yusuf, M., Mkpouto, P., Bashiru, S. & Jaafar, U. (2021). Breakthrough in SARS-COV-2 Monoclonal Antibodies Development. Novel Research in Microbiology Journal, 5(1), 1063-1076.
- Aleruchi, C., Godwin, A.O. & Bashiru, S.E. (2019). Listerial contamination of raw beef and chevon in North-Central Nigeria: Health Implications. IMC Journal of Medical Science, 13(2), 001.

SELECTED CONFERENCE CONTRIBUTIONS

- Sani, B.E., Chuku, A., Obande, G.A. & Uyi, O.G. (2025). Global Trends and Regional Gaps in Mycoremediation: A Systematic Review of Fungal Applications for Environmental Detoxification. 15th Annual Conference of the Mycological Society of Nigeria.
- Bashiru, E.S., Chuku, A. & Orole, O.O. (2023). Genetic Diversity and Antimicrobial Susceptibility of Mucor, Candida and Aspergillus from COVID-19 First Responders in Lafia, Nasarawa State. 13th Annual Conference of the Mycological Society of Nigeria.
- Bashiru, E.S., Chuku, A., Oluwatosin, A.O. & Yusuf, I. (2023). Candidemia in University Students: Detection, Species Distribution and In Vitro Antifungal Susceptibility. 44th Annual Conference of the Nigerian Society for Microbiology.

HONOURS, AWARDS & PROFESSIONAL DEVELOPMENT

- Petroleum Technology Development Fund (PTDF) Fully Funded Overseas PhD Scholarship, 2024.
- Petroleum Technology Development Fund (PTDF) Fully Funded Overseas Scholarship, 2018.
- Dean's Award / Best Graduating Student, Faculty of Science, Federal University of Lafia, 2016; Best Graduating Student, Department of Microbiology.
- Peer Reviewer Certification, Publons Academy, 2021.
- Metagenomics, Technical University of Denmark (Coursera), 2020; Data Management for Clinical Research and Good Clinical Laboratory Practice, The Global Health Network, 2020.

PROFESSIONAL SERVICE & MEMBERSHIPS

- Peer reviewer for scientific journals, including Trends in Sciences and International Journal of Research and Innovation in Applied Science.
- Professional memberships include Applied Microbiology International, American Society for Microbiology, British Society for Antimicrobial Chemotherapy, Nigerian Bioinformatics and Genomics Group, Nigerian Society for Microbiology and Mycological Society of Nigeria.